

Arun Louis

(602) 807-8334 | alouis6@asu.edu | www.linkedin.com/in/arunlouis17

EDUCATION

Master of Science in Data Science, Analytics and Engineering
Arizona State University

Expected May 2026
Tempe, AZ

PROFESSIONAL EXPERIENCE

Automation Developer

August 2021 - July 2024

Ericsson | Bengaluru, India

- Designed and deployed **automated data pipelines** integrating APIs from JIRA, SuccessFactors, and proprietary Ericsson systems, streamlining **data ingestion, validation, and reporting**.
- Built **OCR-powered automation** using UiPath to extract structured data from receipt images, enabling expense classification and compliance verification.
- Processed and analyzed automation-generated datasets using **Excel (PivotTables, Power Query, VLOOKUP)** and **Python (Pandas, NumPy)** for performance metrics, anomaly detection, and client-ready visual reports.
- Managed **Azure-hosted automation workflows**, optimizing scalability and security for enterprise-wide use.

Automation Intern

February 2021 - July 2021

Ericsson | Bengaluru, India

- Developed automation workflows to automate browser, email, and file-processing tasks, reducing manual work by up to 80%.
- Implemented **OCR and rule-based logic** for invoice data extraction, validation, and structured database entry.
- Supported the integration of automation outputs into **analytics dashboards** for operational monitoring and decision-making.

ACADEMIC PROJECTS

Improving LLM Reasoning via Goal-Conditioned Search

August 2025 - Present

Arizona State University | Tempe, Arizona

- Designed a value-guided beam search system that enhanced mathematical reasoning in large language models through goal-conditioned decoding.
- Integrated **Hugging Face Transformers** and modular Python systems for iterative rollout generation and scoring
- Fine-tuned **Qwen-0.5B** and **TinyLlama-1.1B** critic models with **LoRA**, reaching 83% accuracy on partial reasoning evaluation.

Real-Time Object Removal and Seamless Video Restoration

January 2025 – May 2025

Arizona State University | Tempe, Arizona

- Integrated YOLOv8 segmentation, DeepSORT tracking, and LaMa inpainting to remove background objects while preserving scene consistency.
- Added optical-flow background caching and warping for coherent results with camera movement
- Achieved 20 FPS on an RTX 4060; CLI supports live webcam or saved video input.
- Applications: AR video editing, privacy masking, live-stream cleanup

Study Coach (Graph RAG and LLM Fine-Tuning)

December 2024 – March 2025

Arizona State University | Tempe, Arizona

- Developed an intelligent study assistant using **Graph-based Retrieval Augmented Generation (RAG)** for task dependency modeling.
- Fine-tuned LLMs with graph-structured inputs to enhance contextual grounding and personalization
- Designed evaluation pipelines to test response alignment and knowledge coherence
- Delivered a scalable architecture integrating **Python, LangChain**, and fine-tuned LLM modules

TECHNICAL SKILLS

Programming: Python, C, Julia

Data & Analytics: SQL, Tableau, Power BI, NumPy, Pandas, Matplotlib

ML & AI: PyTorch, scikit-learn, Hugging Face, OpenCV, Transformers

Cloud & Automation: Azure, UiPath, Blue Prism, API Integration, OCR

Tools: JIRA, Microsoft Excel, Apache Spark